

interpretation of results as well as on the methods to be used for sampling and repealing Decisions 2002/657/EC and 98/179/EC.

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8

Measurement Uncertainty and Decision

8.1 Framework for Decision-Making

8.1.1 Decision *versus* Uncertainty

The word “uncertainty” has been a major source of inspiration for philosophers, psychologists, economists, and even poets. There is a long list of quotations that play with the word. It originates from the Latin verb *cernere*, which can be translated as discern, distinguish, and finally decide. The past participle of the Latin verb gave *certus* and its opposite *incertus* from which come certain and uncertain. A Latin proverb plays on these words “*amicus certus in re incerta cernitur*,” a “reliable friend can be recognized in unreliable times.” In its most classical sense, uncertainty means not knowing what to do. Etymologically, it is the opposite of knowing how to decide.

On the other hand, the word “decision” is derived from the Latin verb *caedere*, which means to strike, knock down, or cut. To decide is therefore to make a final cut to a question, a controversy, or a situation. Therefore, it seems paradoxical to claim that knowledge of uncertainty improves decision making.

Nevertheless, for a scientist, uncertainty is not synonymous with lack of knowledge, and in Chapters 6 and 7, we have always been careful to speak of measurement uncertainty (MU) rather than *uncertainty* in a narrow and strict sense. It is disappointing that there is such an ambiguity to name what is, in any case, an intrinsic property of a measurement value, namely the manifest dispersion of its replicates, leading to the establishment of a set of possible values for a given result.

Any measurement process has its own variability, and it is essential to accept that the analytical result is never the absolute truth, regardless of the technical improvement that has made this measurement possible. Despite the ambiguity that surrounds a single result, the latter remains fully useful, as demonstrated by the millions of analyses carried out every day, and it allows us to approach this searched-for truth. And the better the uncertainty of measurement is known, the closer the truth.

Chapters 6 and 7 have largely endorsed this affirmation, at least from a statistical point of view. As pointed out in many textbooks, deciding means facing the possibility of making a mistake and various pitfalls and dangers. For example, deciding that a lot is compliant and can be released is exposing oneself to the probability that it is, in fact, noncompliant. But a danger by itself is not the main issue, what really

matters is the probability of its occurrence; this is called the risk. Thus, an avalanche is a big danger with disastrous effects, but it only presents a risk in the mountains; in lowland areas the risk is zero.

Similarly, when a batch is released, if only 1 out of 10,000 units is nonconforming, the risk of receiving a bad unit is not the same as when the frequency is 1 out of 100 units. As explained in Section 6.11, the concept of risk consists of moving from the notion of frequency to that of probability in the same way it is classically achieved in risk analysis or evaluation for insurance companies.

A perfect model of decision-making would be that of the rational actor, who is a pure calculating and optimizing decision-maker. He would integrate all existing information and, on this basis, make the decision that is objectively considered the best. This model is utopian because most of its assumptions are unrealistic. Among other things, the idea that the decision-maker possesses all the information is never true in practice. Decisions are usually based on a relatively small amount of information; this is a context of limited rationality. In regard to risk analysis, there are at least two classical conceptions of uncertainty:

- The first is referred to as fundamental uncertainty and supposes that the future does not follow any predetermined laws of probability.
- The other one is called scientific uncertainty, and it has been our main subject up to now under the name of measurement uncertainty.

When considering the use of analytical data, in many situations, both definitions may coexist, especially when dealing with legal or regulatory issues. The main theories about decision-making under uncertainty come from economists, and there is a large body of literature in this area.

As far as analytical sciences are concerned, the question arose earlier in medical biology laboratories, when the practice became widespread of legally engaging the responsibility of decision-makers and prescribers suspected of having made a wrong decision. But it would be an error to believe that only medical laboratories are concerned. Today, any laboratory providing services can also be confronted with this type of liability and be prosecuted. As an example, when talking about legal liability, jurists distinguish between criminal liability, which is a source of sanctions, and civil or administrative liability, which is a source of compensation.

To make a long story short, the mission of an analytical laboratory providing services consists of carrying out analyses requested by a prescriber, who is generally responsible for interpreting the results. However, it should be noted that in certain specific areas of activity, such as forensic medicine, biomedical analysis, toxicology, and so on, the interpretation of the result is the responsibility of the laboratory itself.

A laboratory is not protected against professional negligence, carelessness, or an unforeseen event: loss of data, accidental disclosure of strategic and confidential information, failure to meet contractual deadlines, etc. In this case, the victim(s) must be compensated for the damage suffered, if it is proven. In concrete terms, this means paying damages and bearing the loss of earnings resulting from the loss of customer confidence. Civil liability insurance is designed to cover these risks and allows the laboratory to be insured against damage caused to third parties during

the laboratory's activities, to property and equipment caused by employees, or to the employees themselves.

8.1.2 Specification Limits and Reference Values

For industrial production lines, control of the process is essential for decision-making, and the use of analytical methods is general. However, this does not mean that all methods must be extremely efficient. The question is rather obtaining results with a level of confidence adjusted to the needs; the notion of *fitness-for-purpose* is consistent with this objective. In fact, it is just necessary to have a method that allows us to control, with a known level of risk, if the technical limit of specification or the regulatory tolerance threshold is exceeded or not.

Estimating the coverage interval of any result provides valuable information to better control the capacity of the result to permit a decision to be taken by reducing the risk of an incorrect decision. It is then easy to affirm that the uncertainty of measurement is an interesting asset for decision-making. If the example of a batch release is regarded, rational control of the risk can be performed in two steps.

- First, the decision maker uses a statistically sound sampling plan to take representative sample units from the lot to be controlled. As explained in Section 8.3, choosing a suitable plan is equivalent to calculating the probability of rejecting a nonconforming lot, i.e. efficiency of the plan. It is calculated *a priori*, by selecting a model that correctly describes the spatial distribution of the analyte in the lot (very often assumed and not measured). Generally, it is recommended to choose an efficiency of 95%, which means that the risk of accepting a not compliant lot is 5%. The idea that 5% of the released batches may be noncompliant is often misunderstood.
- Then, the decision is made on the basis of one or more measurement values obtained from the sampling units collected according to the sampling plan. The decision-maker applies the decision rule that has been chosen. For example, it consists of comparing the measurement values to a specification limit. It is at this stage that MU reduces the risk of accepting a nonconforming lot, especially if the part of the uncertainty due to sampling is included in its calculation.

Finally, the 95% coverage interval of a result is a practical way to take into account the MU.

Figure 8.1 gives a graphical illustration of this proposal. It shows that some results, which would be considered conforming if the MU is not taken into account, are in fact in the rejection interval, and vice-versa, as explained in more details in Section 8.2.2. When comparing a result to a given limit, a classical procedure consists in a significance test, often called null hypothesis testing in statistical literature.

In Section 8.2, it is demonstrated that an alternative and simpler approach to such test is just reporting the coverage interval of the result, without going through the sometimes-delicate calculation of a statistical test. Depending on the field of application there are different manners to set a threshold or limit of specification. For example, it can be:

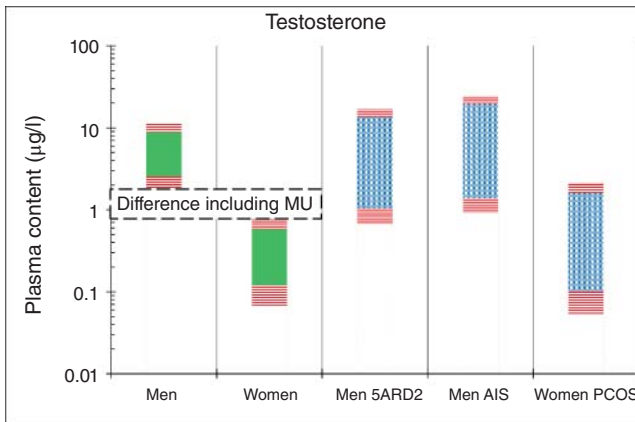


Figure 8.1 Total circulating testosterone reference values for normal and pathological states with associated MU. Logarithmic scale.

- Technical specifications required for the proper realization of industrial objects, such as the nominal content of a drug.
- Environmental or food safety standards defined by official control authorities.
- Declarative values, such as for nutrition labelling.
- Normal and pathological limits used in medical biology.

Moreover, these limits and specifications can be unilateral or bilateral, or defined as lower or upper thresholds. To establish normal or pathological reference values in medicine, several cohorts will be recruited on a diagnostic basis: healthy patients, patients with a certain pathology, others with a second type of pathology, etc.

To obtain acceptable reference values or intervals, it is generally considered that at least 250 individuals per situation (or cohort) are appropriate. Once measurements of the biological biomarker have been collected, the reference values, whether they apply to healthy individuals or those with a pathology, are obtained by identifying the intervals that delimit 95% of the observations of each cohort. The methodology used to determine this interval is variable, and several approaches exist, either empirical or deterministic. In the two-sided case, it is no longer a reference value but a reference interval that contains 95% of the measurements, spreading from 2.5 to 97.5% percentiles.

Figure 8.1 illustrates reference intervals including [2.5%, 97.5%] quantiles for the total circulating testosterone in males and females [1]. The limits between the two groups of normal males and females (plain colors) do not overlap, as expected. Reference interval for each gender is [2.54, 8.90] µg/l for men, and [0.12, 0.58] µg/l for women, respectively. A method of analysis was developed using a LC–MS–MS technique and validated using the accuracy profile procedure [2]. It was possible to compute the relative uncertainty function that is the following power function:

$$UR\% = 0.352.X^{-0.125}$$